Linear Regression Program

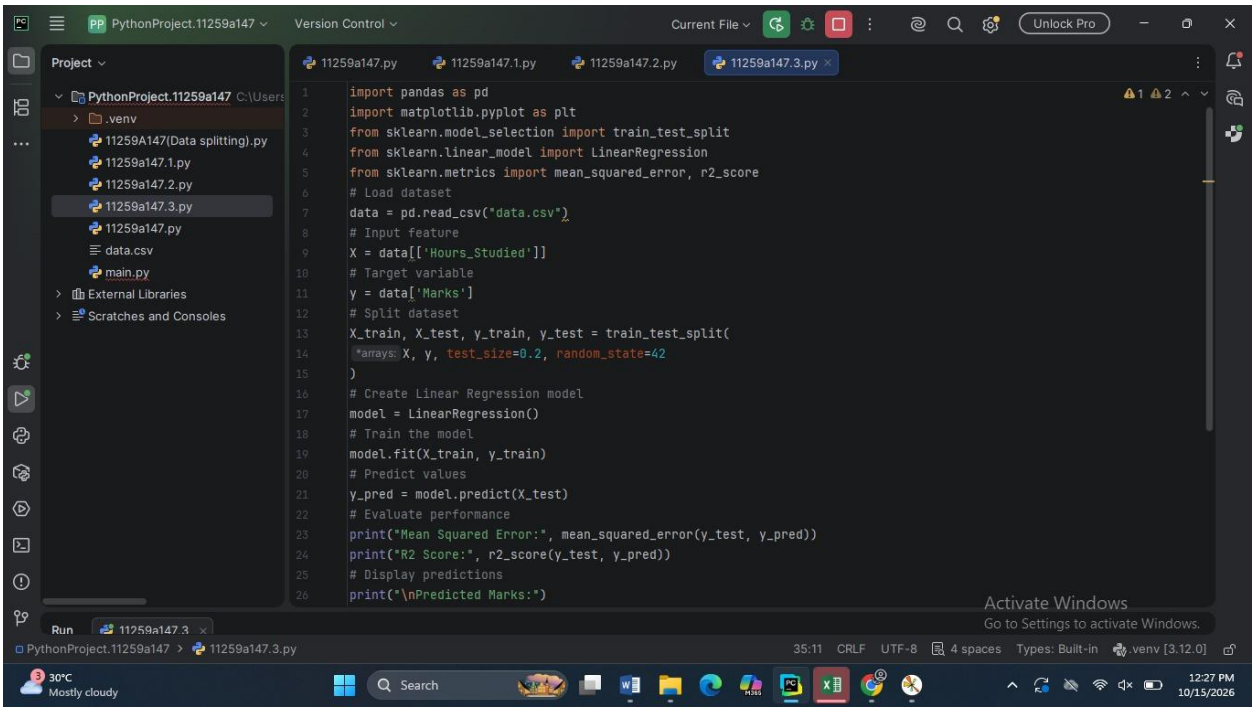
Aim

To write a program that builds a linear regression model on the given dataset, trains it on a training set, predicts values for the test set, and evaluates how well the fitted line explains the data using error metrics and the R^2 score.

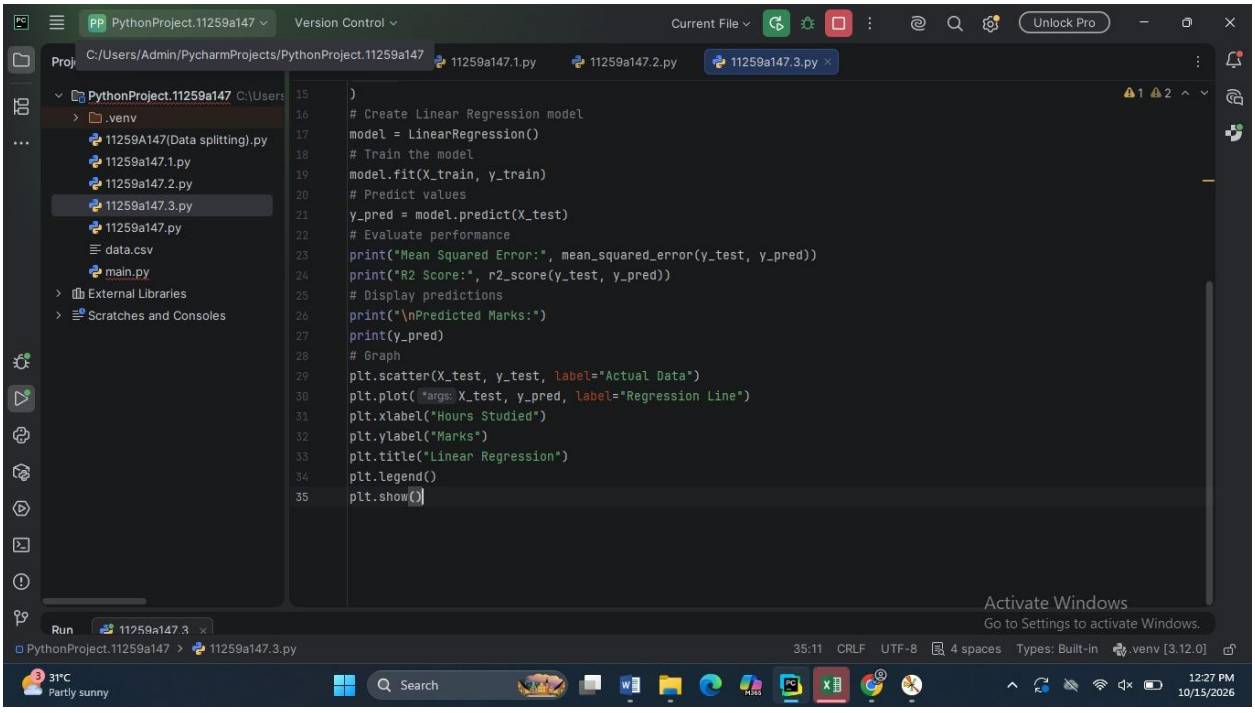
Algorithm

1. Start the program.
2. Import the required libraries (pandas, numpy, matplotlib, train_test_split from sklearn.model_selection, LinearRegression from sklearn.linear_model, and mean_squared_error and r2_score from sklearn.metrics).
3. Load the dataset from the input file.
4. Separate the data into the input feature(s) X and the target variable y.
5. Split the data into training and testing sets using train_test_split (for example 80% training and 20 % testing), with a fixed random_state.
6. Create the LinearRegression model object.
7. Train the model by fitting it on the training data (X_train, y_train).
8. Display the learned slope (coefficient) and intercept.
9. Predict the target values for the test set (X_test).
10. Calculate the Mean Squared Error and the R^2 score by comparing the predicted values with the actual values (y_test).
11. Display the predicted values, the MSE and the R^2 score.
12. Plot the actual data points and the fitted regression line (optional).
13. Stop the program.

Program:



```
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score
# Load dataset
data = pd.read_csv("data.csv")
# Input feature
X = data[['Hours_Studied']]
# Target variable
y = data['Marks']
# Split dataset
X_train, X_test, y_train, y_test = train_test_split(
    *arrays: X, y, test_size=0.2, random_state=42
)
# Create Linear Regression model
model = LinearRegression()
# Train the model
model.fit(X_train, y_train)
# Predict values
y_pred = model.predict(X_test)
# Evaluate performance
print("Mean Squared Error:", mean_squared_error(y_test, y_pred))
print("R2 Score:", r2_score(y_test, y_pred))
# Display predictions
print("\nPredicted Marks:")
```



```
)
# Create Linear Regression model
model = LinearRegression()
# Train the model
model.fit(X_train, y_train)
# Predict values
y_pred = model.predict(X_test)
# Evaluate performance
print("Mean Squared Error:", mean_squared_error(y_test, y_pred))
print("R2 Score:", r2_score(y_test, y_pred))
# Display predictions
print("\nPredicted Marks:")
print(y_pred)
# Graph
plt.scatter(X_test, y_test, label="Actual Data")
plt.plot( *args: X_test, y_pred, label="Regression Line")
plt.xlabel("Hours Studied")
plt.ylabel("Marks")
plt.title("Linear Regression")
plt.legend()
plt.show()
```

Result:



Graph:

